



United International University

Department of Computer Science & Engineering

CSE 3715: Data Communication

Class Test 01

Duration: 40 Minutes

Total Marks: 20

ID:

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Short Question Answers ($4 \times 2.5 = 10$ Marks)

1. Why does the transport layer need error detection even though the data link layer already provides it?
2. Why does the network layer use IP address while data link layer uses MAC address?
3. Why are port numbers required in the transport layer during end-to-end communication?
4. Why is data segmentation performed in the transport layer before transmission?

Simulation (1 × 10 = 10 Marks)

- Based on the given topology in the figure (figure 01) and the corresponding IP, MAC, and Port addresses in the table (table 01), complete the packet content table (table 02) for each hop along the path.

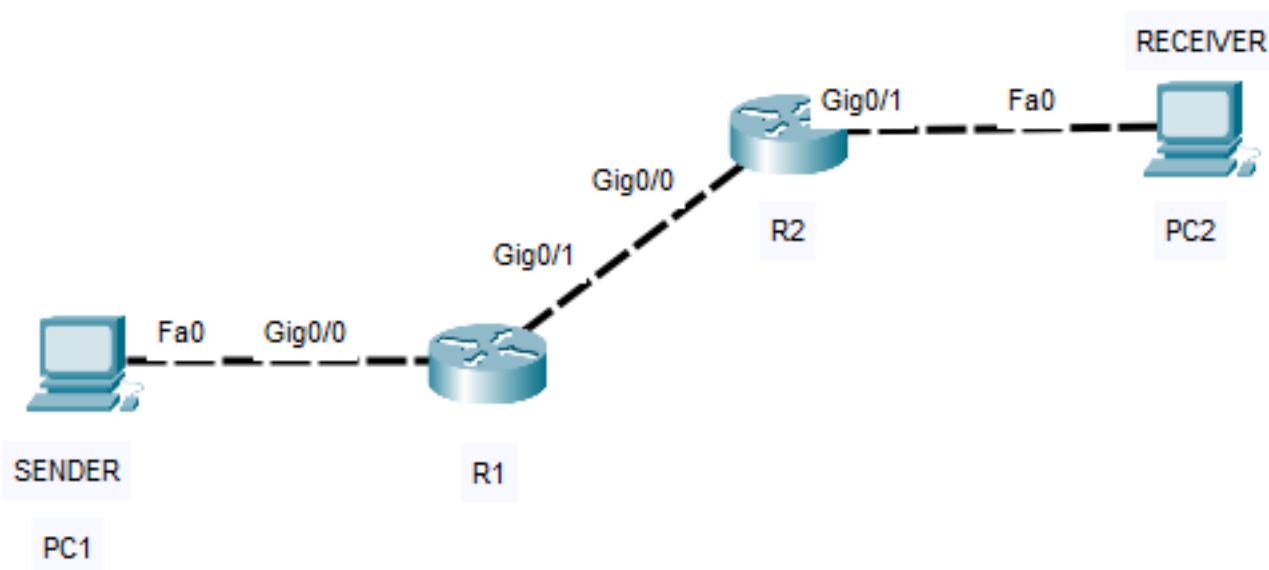


Figure 1: Physical Topology

Device	Interface	IP Address	MAC Address	Port Number (TCP)
PC1	G0/0	192.168.1.2	3C:52:82:1B:9F:47	80 (HTTP)
R1	G0/0	192.168.1.1	00:1C:42:7B:3D:F0	-
R1	G0/1	10.0.0.1	00:E0:4C:23:9A:5B	-
R2	G0/1	10.0.0.2	00:50:56:C0:00:08	-
R2	G0/0	192.168.2.1	00:25:96:FF:FE:12	-
PC2	G0/0	192.168.2.2	B8:27:EB:45:23:1A	22 (SSH)

Table 1: Addressing Table

Hop	Destination MAC Address	Source MAC Address	Source IP Address	Destination IP Address	Source Port	Dest. Port
PC1 → R1						
R1 → R2						
R2 → PC2						

Table 2: Packet Contents